Enforcing Hidden-State Policies in Native Private Aggregation

***Abstract*—Private heatmap aggregation over a predeclared client cohort requires authorizing hidden-state transitions without exposing client coordinates to the admission authority. Composing zero-knowledge state proofs with secret-shared aggregation leaves two semantic gaps: an individually valid share vector can encode a cell different from the state-authorized cell, and aggregators can agree on a batch that omits accepted reports or exposes a distinguishing subset. Stateful History-to-Payload Closure (SHTPC) binds hidden-state authorization, retained aggregator shares and complete-cohort first exposure. Within the one-hot histogram domain, a vector's linear index moment uniquely identifies its selected cell; aggregators sign masked-moment receipts over retained shares so admission verifies cell equality without encoding PRG seed expansion in the application circuit. Atomic admission serializes state transitions, and durable reservation locks the complete cohort before any honest aggregate share becomes decryptable. With honest services, SHTPC preserves accepted-history integrity across windows and crash recovery; against an active aggregator colluding with the Collector, it hides trajectories up to public metadata and equal declared complete-cohort histograms across single-release and predeclared multi-window executions. At 10,001 bins, the standalone binding circuit uses 5,779 constraints (versus 79,002,390 for in-circuit seed expansion), accelerates transition proving by 89.8–89.9× over Poseidon vector binding, and adds 1.44× batch latency (1.38× per session) over plaintext trusted generation in 100-client Janus runs while blocking cell mismatches and incomplete-cohort exposure.**

1. Introduction

A managed fleet with predeclared participants and coordinated sampling uses cohort heatmaps to detect congestion and dispatch vehicles without revealing individual driver trajectories [1–3]. Even when aggregation arithmetic is exact, semantic and membership mismatches distort operational decisions: redirecting reports away from their history-authorized cells fabricates hotspots—inflating a T-Drive modal cell count from 3 to 22–23 when 20 of 100 reports are substituted in offline replay (Section 5.2)—whereas omitting accepted reports suppresses true demand (Figure 1).

The application combines an admission authority that authorizes transitions without reading client coordinates with two aggregation roles that execute a native secret-shared aggregation protocol [4–6]. A Collector consumes the resulting cohort statistics. Private aggregation validates individual inputs and combines them without revealing each input. The remaining task is to connect the authorized transitions, retained contributions and release membership to the same accepted history.

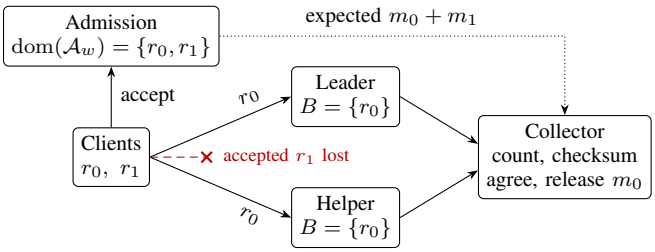


Figure 1. Independent composition can omit a report already accepted by application admission on both role channels. Here $\mathrm{dom}(\mathcal{A}_w)$ denotes the set of report IDs accepted by admission for window w, and B is each role's processed native batch.

Existing protocols validate private inputs [7–10], enforce inclusion or verify aggregates [11–13], and maintain private aggregate state [14]. Suppose a state proof authorizes cell p, while a native-valid report encodes cell $q \neq p$. If the checks examine these objects independently, they accept the cross-spliced pair. A common identifier or signed body associates the objects but does not establish semantic equality. A hiding commitment binds its opening, but an additional relation connects that opening to the native output actually retained. Because independent checks form a product set of locally valid objects, they inevitably accept cross-spliced pairs (formalized in Appendix A).

State freshness and release membership create two further interfaces. A proof for predecessor c remains valid after a competing successor commits, so proof verification does not authorize the current transition. Separately, an accepted report can be dropped on both receiver channels. The receivers then agree on a valid but incomplete batch, as Figure 1 shows. Both interfaces therefore depend on information beyond independent proof and native-input checks.

Completeness also has a privacy deadline. Two legal histories can swap two owners' positive cells while retaining the same public metadata and complete heatmap. A subset containing only one of those owners has different aggregates. Once the first honest share is decryptable, a corrupted aggregation role and the Collector can distinguish the histories. A later publication refusal cannot retract the exposure. Proposition 1 formalizes this distinction. Accepted-history closure connects each admitted claim to its retained contribution and legal state transition, and reconciles the eligible set before first exposure.

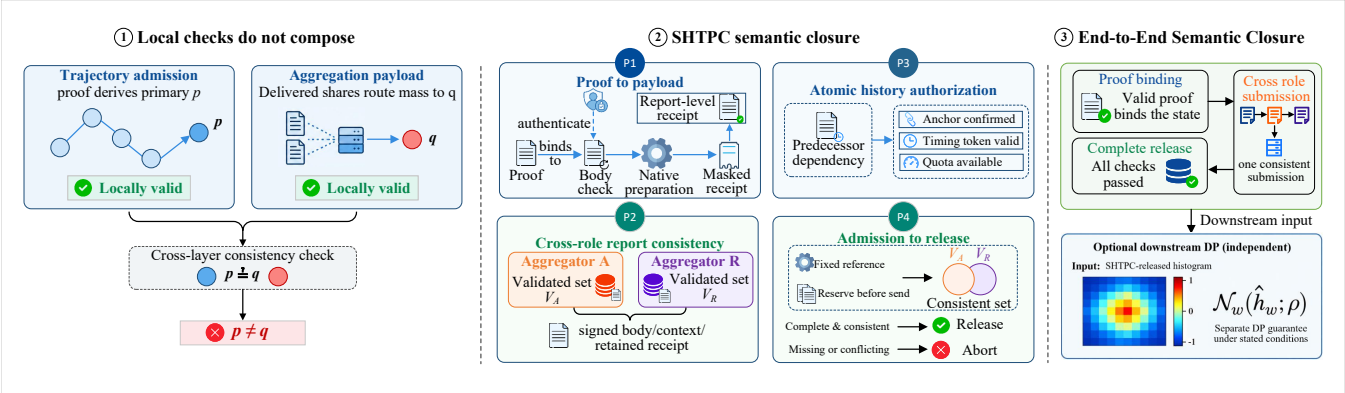


Figure 2. SHTPC connects report meaning, accepted history and complete release through four semantic boundaries.

Stateful History-to-Payload Closure (SHTPC) binds each hidden-state proof to the share vector retained by the Aggregators without reproducing native seed expansion inside the application circuit. Native histogram validity restricts the reconstructed contribution to a one-hot vector with a sentinel. Within this domain, its linear index moment identifies the selected cell. Each role computes a masked receipt from its retained output, and admission compares the paired receipts with the proof-bound masked index. The application circuit is therefore independent of histogram width D, while native vector processing remains $O(D)$.

The bridge introduces receipts and retry feedback that require joint analysis. Authenticated source registration binds private processing to its registered body, and joint simulation accounts for the resulting transcript. Atomic admission advances the current history, while durable reservation fixes the complete cohort before exposure. Figures 2 and 3 show their execution order. Integrity assumes honest services facing malicious clients and transport faults. Privacy permits one actively corrupted aggregator colluding with the Collector. The single-release and predeclared multi-window analyses require common metadata and equal declared histograms for each permitted complete cohort.

SHTPC contributes an accepted-history contract, a compact native-output binding and an analysis of their composition with private aggregation.

1) *Masked-moment binding.* Native one-hot validity and masked receipts from retained outputs bind a hidden-state proof to the actual contribution without reproducing native seed expansion in the application circuit. The standalone relation has 5,779 constraints at 10,001 bins, versus 79,002,390 for direct seed binding.
2) *Accepted-history closure.* A security contract connects each admitted claim, retained contribution and legal state transition to complete eligible cohort membership at first exposure. A separation example shows why publication refusal alone is too late. The construction maintains these connections across retries and recovery under honest service execution.
3) *Joint privacy of receipt feedback.* Masked receipts, rejection feedback and retries are jointly simulatable with the native transcript against an active aggregator colluding with the Collector. The exposure-time interface uses no aggregate before the first honest share becomes decryptable and one permitted complete-cohort output at that boundary. Simulation distinguishes malicious sources' declarations from their actual retained outputs. Native controls exercise these interfaces, and component and lifecycle measurements report their costs.

2. Model and Security Contract

2.1. Deployment and Trust Model

Enrolled clients report under stable pseudonymous identities. An admission authority verifies hidden state proofs and maintains accepted history without receiving coordinate openings or role decryption keys. Two Aggregators, acting as the standard Leader and Helper in the underlying DAP protocol [4] (algebraically roles $S \in \{A, B\}$, or corrupted c and honest h in privacy analysis), prepare input shares and retain aggregation outputs. The Collector holds the aggregate-share HPKE key, and its Decoder verifies release records and reconstructs the heatmap. The authority-side Finalizer authenticates the frozen reference and its completion. Its certificate joins that reference to both roles' matching attestations. A durable admission proxy belongs to the same trust domain and delivers signed decisions. Each role durably reserves its own share before export.

Integrity and privacy operate under distinct corruption models. For integrity, clients and network transport are malicious, while admission, both Aggregators, Finalizer, Collector and Decoder follow the protocol using serialized durable storage; this fits a managed fleet enforcing admission and release rules without granting the authority plaintext coordinates. For privacy, one Aggregator may be actively corrupted in collusion with the Collector, and the adversary may additionally observe the honest authority's business records, receipts, diagnostics and backups (excluding its signing key). Because an active Aggregator can forge shares or abort, this corruption model targets privacy rather than output integrity; a corrupted admission authority is

outside both models. Crucially, forged receipts cannot serve as a probing oracle. Admission acceptance for an honest report depends solely on the corrupt role's own fixed original share and mask, leaking no honest coordinates (Section 4.2).

Privacy challenges use legal histories with common public structure μ, including identities, task and key, windows, step and anchor references, eligibility, contribution bits, packet rules and the delivery plan. Proofs, ciphertexts, commitments, receipts, errors, aborts and retries are simulated jointly. The authority has no aggregate interface. Definition 6 gives the deployed single-release contract, with one visible cohort per task and key. Theorem 5 extends the analysis to predeclared windows in the same task and key namespace. Reports and retained outputs each belong to one window. Later windows can reference earlier accepted states as predecessors or anchors. Privacy bounds protocol leakage beyond the permitted histograms and μ. Activity, contribution bits and cadence remain public, and exact outputs can support inference from auxiliary knowledge.

All guarantees assume correctly sampled public parameters with unknown trapdoors, reviewed fixed-width arithmetic, authenticated confidential registration and durable execution boundaries. Enrollment authenticates pseudonymous identities rather than physical GPS locations or Sybil resistance, continuity is enforced over authenticated slots rather than unverified elapsed time, and authority equivocation lies outside this scope.

2.2. Reports, Histories and Cohorts

Reports move through three distinct units of progress. A sampling slot assigns a measurement to time, an accepted step counts a committed transition, and a release window groups eligible reports. Rejected attempts do not advance the accepted history.

Each report is projected and quantized under fixed public parameters, yielding a hidden vector $\mathbf{x}_r$. A $W \times H$ grid with cell width s selects

$$p_r = \mathsf{Grid}(\mathbf{x}_r) = W \left\lfloor \frac{y_r}{s} \right\rfloor + \left\lfloor \frac{x_r}{s} \right\rfloor, \tag{1}$$

where $D = WH$. The evaluated profile uses $s = 100$ and $W = H = 100$. The replay projection, rounding and boundary rules appear in Appendix A.

All record maps are keyed by report identifier, tracking progress across the three lifecycle stages in Table 1. First, during *registration and locking*, clients register report cores before preparation feedback, forming the complete registered history map P_w and its eligible projection R_w. Warmups enter log P_w to advance state counters but are excluded from eligible map R_w, whereas eligible zero-bit records ($b_r = 0$) remain in R_w despite adding no mass. Let $I_w = \mathrm{dom}(R_w)$; honest report identifiers form I_{hon}, partitioned as $I_w^{\mathrm{hon}} = I_w \cap I_{\mathrm{hon}}$ and $I_w^{\mathrm{adv}} = I_w \setminus I_{\mathrm{hon}}$. Projection π_{reg} preserves registered fields while removing later receipts, commit decisions, and retained-output links; map equality compares keys and values. The concurrent log is $\mathcal{L}$, and H is a legal sequential history. Second, during

TABLE 1. RECORD MAPS AT SUCCESSIVE EXECUTION STAGES.

Stage	All history	Eligible
Registered and locked	P_w	R_w
Admission committed	L_w	$\mathcal{A}_w$
Role S applied	$L_{S,w}$	$M_{S,w}$

admission commitment, reports that pass zero-knowledge transition proofs, masked receipts, and atomic history checks are committed, yielding committed log L_w and eligible map $\mathcal{A}_w$. Third, during *role application*, roles $S \in \{A, B\}$ process committed decisions into applied maps $L_{S,w}$ and $M_{S,w}$. For each applied report r, role S retains share $v_{S,r} \in \mathbb{F}_q^{D+1}$ with sentinel; its reserved sum $\mathbf{a}_{S,w}$ yields the reconstructed D-cell heatmap $\hat{\mathbf{h}}_w$ upon decoding.

For report r, the common admitted object is

$$A_r = (\mathsf{rid}_r, \eta_r, \mathsf{stmt}_r, \mathsf{pay}_r, c_{r-1}, c_r, w, b_r). \tag{2}$$

The context η_r binds identity, window and policy. The remaining fields identify the proof statement, bound channel packages, predecessor, successor and contribution bit. A complete signed body, two retained-output receipts and one commit decision authenticate this object in the native realization (Appendix B).

Strict release produces the complete eligible cohort or refuses, serving applications that require coordinated sampling and can tolerate delayed batch release. Execution separates failures across three distinct phases: (1) *Pre-lock registration failure*, where a client fails to submit a complete signed body or assign a required contiguous slot before the registration lock boundary; (2) *Post-registration validation stall*, where an installed body contains an invalid proof, references an uncommitted anchor, or violates quota; and (3) *Post-commit transport failure*, where network disconnection occurs after an admission decision is durably committed. Critically, once a client's complete request is registered and committed, subsequent vehicle disconnection does not halt cohort processing: server-side durable redelivery and idempotent retry continue role application and export autonomously without requiring continuous client online presence. A "permanently missing member" specifically denotes a client failing Phase (1) registration or Phase (2) proof validation. If any member is permanently missing, release is strictly refused; SHTPC preserves complete-cohort integrity and prohibits dynamic member dropouts or publishing over the receiver intersection set. Table 2 gives its window lifecycle.

2.3. Accepted-History Closure

Four obligations share A_r. P1 connects the proved claim to the consumed package by

$$\mathsf{Contrib}(\mathsf{pay}_{r,\mathrm{consumed}}) = b_r \mathbf{e}_{p_r}. \tag{3}$$

P2 requires both roles to consume channels of that same object. P3 gives each accepted object one atomic transition with its quota charge,

$$\mathrm{State}_{r-1} \to \mathrm{State}_r. \tag{4}$$

TABLE 2. NATIVE WINDOW LIFECYCLE AND SCOPE.

Stage	Contract
Register	All predeclared UID and contiguous step slots, including warmups and eligible zeros, need full pending bodies.
Lock	At the registration boundary, the complete graph starts at actual heads. Both roles install the lock.
Process	Current-head permits precede preparation and commit. Both roles apply decisions.
Expose	Complete records are reserved before the first share. Retries use the same snapshot.
Abort	Partial commits and quota persist. Later configured slots and new bodies start at actual heads.

P4 requires a successful release to contain exactly the eligible committed records and the complete locked reference,

$$\pi_{\mathrm{reg}}(\mathcal{A}_w) = R_w. \tag{5}$$

These obligations organize the successful-trace and exposure-prefix definitions below. Theorem 1 establishes their composition under the stated integrity premises.

Definition 1 (Legal sequential history)**.** A sequence $H = (A_1, \ldots, A_n)$ with unique identifiers is legal if replay from enrolled genesis enables every transition. Each step has an authorized owner and context, names the current head and policy-selected accepted anchor, and satisfies the application relation on opened states. It atomically appends the successor, advances the head and accepted-step counter, and charges quota without changing other owners. Rejected and uncommitted attempts add no step.

Definition 2 (History refinement)**.** For a finite prefix, $\mathcal{L} \preceq H$ when committed admission operations bijectively map to a legal H, preserving every A_r field and the order of linearization points between invocation and response. Committed pending operations are completed at the prefix end, while uncommitted attempts are discarded. Retries of one durable decision represent one operation, and distinct committed effects remain distinct.

Definition 3 (Successful-trace closure)**.** Fix a transition policy, additive contribution map $r \mapsto m_r$ and eligibility rule. A successful release has closure if some legal H satisfies $\mathcal{L} \preceq H$. Let $E_w(H)$ be the set of records in the complete accepted history H that belong to window w and satisfy the eligibility rule. Their identifiers equal $\mathrm{dom}(\mathcal{A}_w)$, and each identifies one common admitted object across proof, packages, transition and certificate. Each role's application count $N_{S,w}(r)$ satisfies $N_{S,w}(r) = \mathbf{1}[r \in \mathrm{dom}(\mathcal{A}_w)]$, and its reserved sum uses exactly those retained outputs. The pre-noise output is $\sum_{r \in \mathrm{dom}(\mathcal{A}_w)} m_r$. For heatmaps, $m_r = b_r \mathbf{e}_{\mathsf{Grid}(\mathbf{x}_r)}$. Eligibility is independent of bit value.

Definition 4 (Exposure-prefix safety)**.** Every decryptable honest share comes from a prior durable reservation satisfying

$$\pi_{\mathrm{reg}}(L_{S,w}) = P_w, \qquad \pi_{\mathrm{reg}}(M_{S,w}) = R_w = \pi_{\mathrm{reg}}(\mathcal{A}_w), \qquad \mathbf{a}_{S,w} = \sum_{r \in I_w} v_{S,r}. \tag{6}$$

Retries preserve its membership and plaintext. Successful-trace closure also requires correct final reconstruction.

TABLE 3. MISSING INTERFACES, COUNTEREXAMPLES AND SUPPORTING CHECKS.

Correct mechanisms	Open interface	Counterexample and check
Proof and native input validity	Proved value equals the retained output	Cross-splicing counterexample (App. A). Native mismatch rejected (Sec. 5.3)
Report identifier and signatures	Authenticated source before private processing	Ciphertext-transplant attack. Native guard-off leak and guard-on rejection (Sec. 5.4)
Proof verification	Head and quota recheck at commit	Stale-head or duplicate-retry counterexample. Native trace and bounded model (Sec. 5.5)
Admission and set checking	Complete set fixed before the first decryptable share	Subset separation (Prop. 1). Native seven-of-eight exposure blocked (Sec. 5.4)
Set checking	Durable reservation before send	Post-send crash counterexample. Persistence premise (Sec. 3.6)

Table 3 identifies missing interfaces and separates their counterexamples from native controls, bounded models and execution premises. Appendix A formalizes cross-splicing and identical-receiver-view arguments.

3. SHTPC Design

SHTPC realizes P1–P4 by connecting hidden-state authorization to retained aggregation outputs and complete first exposure.

3.1. Design Overview

Anchor Distance Window Continuity (ADWC) bounds each hidden move relative to the last accepted position and a policy-selected historical anchor. The client proves ADWC and grid membership over hidden states. Native validity and masked-moment receipts connect the proved cell to the outputs retained by the aggregation roles. Admission checks these receipts and authorizes their use against the current accepted history. Figure 3 shows how this connection reaches a complete release.

3.2. End-to-End Workflow

Enrollment fixes owner keys, a proved hidden genesis and the task configuration. Each window then proceeds through five steps.

1) *Lock.* Clients register complete signed bodies with admission. The authority verifies the candidate graph and locks P_w and R_w before private preparation or receipt feedback.
2) *Retain.* Both roles install the lock. A permit authorizes preparation against the current head and anchor. Each role prepares its input, retains its native output and signs a masked-moment receipt.
3) *Commit.* Admission checks both receipts and their masked-moment equality. A transaction rechecks the actual head, anchor and quota, then stores the successor, history and signed decision together.
4) *Apply.* A durable proxy delivers the decision to each role. Each role applies it once and updates its retained history and eligible map. Warmups advance history without entering the eligible map.
5) *Reserve.* Each role checks its applied records against the frozen reference and verifies the retained-output links. It durably reserves the complete sum and selector before sending its share. Retries use the same snapshot.

Preparation creates a retained output, while admission authorizes its use in aggregation.

3.3. Binding Hidden State to Native Outputs

P1 binds the state-authorized cell to the retained native output without encoding native PRG seed expansion inside the zero-knowledge circuit. Under native private aggregation, the underlying verifiable aggregation layer guarantees one-hot validity of the reconstructed vector ([5, Section 7.4.4]): for roles A and B retaining shares $v_A, v_B \in \mathbb{F}_q^{D+1}$, the reconstructed output satisfies $v = v_A + v_B = \mathbf{e}_y$ for a unique unit coordinate $y \in [0, D]$.

Over this one-hot domain, the linear index moment $\sum_{j=0}^{D} jv[j] = y$ uniquely identifies the selected index y (for example, in a three-cell vector $(0, 0, 1)$, $0{\cdot}0{+}1{\cdot}0{+}2{\cdot}1 = 2$). In native field $\mathbb{F}_q$ with $D < q$, let a state-derived primary $p < D$ and public bit b authorize target index $x = bp + (1 - b)D$, with sentinel D. The client samples private uniform masks $r_A, r_B \in \mathbb{F}_q$, commits to their openings, and encrypts r_S for role $S \in \{A, B\}$. The ZK proof binds x and the masks to the public target T:

$$T = x + r_A + r_B \pmod q. \tag{7}$$

Each correct role computes a local masked moment from its retained output:

$$m_S = \sum_{j=0}^{D} jv_S[j] + r_S \pmod q. \tag{8}$$

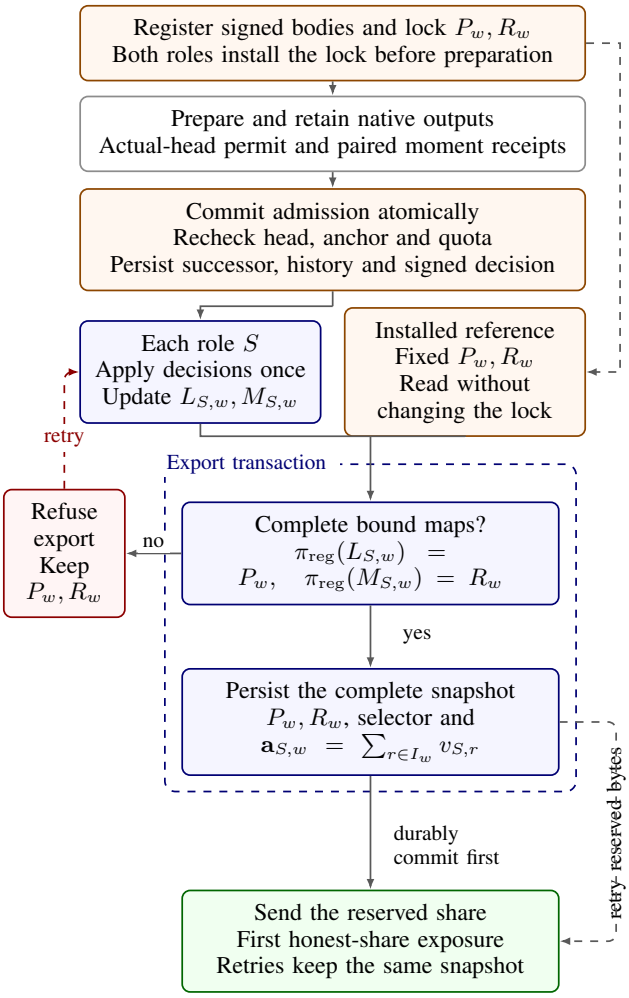


Figure 3. SHTPC reserves the complete eligible history before share exposure.

Because moments are linear, summing the two receipts causes the masks to cancel:

$$\begin{aligned} m_A + m_B &= \sum_{j=0}^{D} j(v_A[j] + v_B[j]) + r_A + r_B \\ &= y + r_A + r_B \pmod q. \end{aligned} \tag{9}$$

Consequently, checking $m_A + m_B = T$ directly tests $y = x \pmod q$. Lemma 1 shows that acceptance forces the reconstructed contribution to equal $b\mathbf{e}_p + (1 - b)\mathbf{e}_D$, recovering $b\mathbf{e}_p$ upon sentinel removal. Table 4 compares this approach against generic vector binding and direct seed binding.

Transition circuit relation (ADWC and grid). The public statement binds the state commitments, report context and masked index. Its witness opens the state, ticket, mask and receipt commitments. The relation binds the payload witness's cell and remainder to the successor and enforces $p = \mathsf{Grid}(\mathbf{x}_r)$, $\|\mathbf{x}_r - \mathbf{x}_{\mathrm{prev}}\|^2 \le \tau_u^2 \Delta t_t^2$ and $\|\mathbf{x}_r - \mathbf{a}_t\|^2 \le \min\{\tau_u^2 (\Delta T_t^{(K)})^2, c_{\mathrm{policy}}^2\}$. Here $\mathbf{a}_t$ is the policy-selected accepted anchor, τ_u is the speed tier and c_{policy} is the anchor

TABLE 4. BINDING CHOICES AND THEIR CONNECTION TO THE NATIVE CONTRIBUTION.

Approach	Certified connection	Additional work or condition
Association only	Common report or committed value	Retained-output equality still needed
Direct seed binding	Seed expansion and selected vector	Native sharding in the circuit
Generic vector binding	Retained shares equal certified vector	Retained-share vector commitment
SHTPC (Proposed)	Certified index equals retained index	Native one-hot validity

cap. Admission checks that the referenced predecessor and anchor are the actual accepted ones. The same grid and distance constraints apply when $b = 0$. Appendix A gives the parameters, and the artifact's fixed transition circuit gives the ordered public-signal interface (Section 8).

Cross-field canonical encoding ($\mathbb{F}_q$ vs. p_{proof}). To bridge native field $\mathbb{F}_q$ and proof scalar field p_{proof}, canonical scalars enforce $x + r_A + r_B = T + \chi q$ with carry $\chi \in \{0, 1, 2\}$ over integers, preventing wrap-around between disparate field moduli (Appendix B). State commitments retain birth domains and references, and admission verifies the masked-moment sum without receiving coordinate openings or Aggregator decryption keys.

Complexity and idempotent preparation. Because moment checking operates on scalars while share evaluation scans $D + 1$ buckets, the application circuit size is independent of histogram width D, whereas native vector processing remains $O(D)$. Separating share preparation from accumulation ensures that each signed receipt authenticates the package context and retained share, with exact retries replaying the same output and receipt.

The compact binding relation relies on linear injectivity over the one-hot domain: a single first-order index moment is injective specifically on unit vectors $\mathbf{e}_y$, but is not injective on general vectors (for instance, $(0, 2, 0)$ and $(1, 0, 1)$ yield the identical moment 2). Extending this moment-cancellation bridge to other stateful applications requires domain-appropriate mappings: multi-dimensional spatial or categorical outputs may require multiple higher-order moments or component-wise binding, and weighted contributions require jointly binding weights and indices (e.g., certifying both $\sum w_j$ and $\sum j w_j$). Stateful applications such as bounded cumulative usage can adopt this approach whenever their reconstructed output domains admit an injective moment relation.

3.4. Protecting the Feedback Interface

Encryption does not authenticate an input's owner. An attacker can transplant an honest ciphertext and substitute a known mask, causing an honest receipt to reveal the cell before admission rejects the report. This feedback arises from the new bridge even when the proof and native protocol are secure in isolation.

SHTPC allocates an owner identifier to one complete signed body and registers it before private processing. Each ingress authenticates the consumed metadata, ciphertexts, sidecars, installed lock and permit against that body. Both retained sessions and receipts therefore refer to the same admitted object, realizing P2. Source registration, origin locking, and fixed original shares prevent an active adversary from splicing components across reports or adapting receipts to probe honest state. Because the original share and mask are permanently locked to the registered body, and retry/abort states are strictly serialized, adaptive feedback cannot alter the receipt passing condition.

Recovery retains the same source and native state. The Helper journals its first protected input and response before sending. Identical retries replay those bytes, while changed inputs or new jobs cannot reopen that branch. The Leader's lease and transaction fence permit at most one committed transition. Uncommitted computation can receive new peer feedback, which the joint privacy analysis includes explicitly.

3.5. Atomic History Authorization

A transition proof can remain valid after another successor commits. Admission therefore verifies the proof before the transaction, then atomically rechecks the actual head, anchor and quota and stores the successor, history and signed decision. Both retained-output checks precede this linearization point. Preparation, admission and role application use separate databases connected by that decision and its output links.

For stable identity u, task and window w, durable quota $q_{u,w} \in \{0, 1\}$ is shared across credential epochs. With accepted round t and $N_{\mathrm{warm}} = \max(K, 2)$,

$$\begin{gathered} \gamma_t = b_t \mathbf{1}[t > N_{\mathrm{warm}}], \\ q_{u,w} + \gamma_t \le 1, \qquad q_{u,w} \leftarrow q_{u,w} + \gamma_t. \end{gathered} \tag{10}$$

Commit induction gives

$$q_{u,w} = \sum_{\substack{r \in \mathcal{A}_w \\ \mathsf{uid}(r) = u}} b_r \le 1. \tag{11}$$

Warmups remain in the transition log without charge. Eligible zero-bit updates remain members and can advance state after quota is spent.

History is keyed by stable $(\mathsf{task}, \mathsf{uid})$, with credential, revision, window, receipt and budget checks inside the transaction. Credential epochs and storage revisions do not replace accepted-step counters or sampling slots. The proxy queues exact decisions durably before delivery. Role databases apply each decision once, including out-of-order delivery. Retries return that decision, and credential rotation preserves history and spent quota.

3.6. Complete Exposure and Recovery

P4 fixes membership before preparation feedback can influence it. Admission locks the verified candidate graph

and its eligible projection at the public registration boundary. The graph includes warmups and eligible zeros together with their predecessor and anchor references. Preparation errors and receipts cannot select a smaller cohort or replace a locked body. Export by role S requires

$$\pi_{\mathrm{reg}}(L_{S,w}) = P_w, \qquad \pi_{\mathrm{reg}}(M_{S,w}) = R_w. \tag{12}$$

These equalities compare the frozen registration cores, including context and body. The role separately checks each actual retained-output link against its authenticated receipt and commit decision.

In one export transaction, the role verifies Eq. (12) and reserves the selector, record map, retained-output links, and exact cohort sum (under the strict profile) before sending its share. Because role application is monotone ($0 \to 1$), this atomic check refines Definition 3 by ensuring all locked members are applied before exposure (Appendix A). Persisting the resulting snapshot before transmission ensures that re-exports idempotently re-emit the identical complete sum without altering membership. Warmup completion is checked through $L_{S,w}$, and eligible zeros remain in $M_{S,w}$; numerical equality alone cannot establish this membership because an omitted eligible zero has no displayed mass. Responses and re-encrypted retries read the same durable snapshot, so recovery cannot change its plaintext or membership. The Finalizer authenticates the frozen reference and both roles' matching attestations. The Decoder checks that certificate and the reserved records before reconstruction.

After locking, any member failing registration or validation holds or aborts the cohort without silent exclusion or intersection release (Section 2.2). Conversely, committed decisions survive client disconnection via autonomous server-side redelivery without re-upload. Collection failure preserves the first exposure and reserved snapshot. Pre-exposure abort retains partial commits, heads and quota for continuation as in Table 2. For example, owners left at steps 2 and 1 resume at 3 and 2 with new bodies referencing those heads. An incompatible schedule cannot lock. Configuration remains bound to the database. Theorem 5 gives the privacy conditions for predeclared windows.

For a finite fully committed lock, eventual service and channel recovery with fair retries and no deadline stop gives complete release. Each successful apply removes one of at most $2|P_w|$ pending role-record pairs. Permanent client refusal and incomplete pre-lock registration fall outside these progress conditions. Admission caps bound accepted work and exact retries consume no new slot. The native profile uses $B_u = 16$ per owner and window.

4. Security Analysis

The integrity argument connects three steps: extraction identifies the certified witness, the moment check binds it to the retained contribution, and atomic admission with durable reservation extends this binding to first exposure. The privacy argument separately simulates the receipts and native transcript, using the permitted complete-cohort histogram at first exposure. Full symbolic security ledgers, concrete hybrid sequences and reduction proofs are provided in the anonymous artifact repository (indexed in Section 8).

TABLE 5. SECURITY PROPERTIES AND OBSERVER VIEWS RELY ON DISTINCT MECHANISMS AND PREMISES.

Property or observer	Main mechanism	Premise or permitted output
Legal hidden witness	Groth16 [15] extraction, Pedersen [16] binding	Unknown trapdoors and typed generic-group extraction
Proved cell is counted	One-hot validity, masked moments	Correct retained outputs and $D < q$
Legal history step	Authentication, commit induction	Atomic head and quota update with durability
Complete first exposure	Lock, apply and reserve	Lock before feedback and persist before send
Authority privacy	Blinding, ZK, HPKE and receipt simulation	Common metadata and no aggregate interface
Role and Collector privacy	Full-verifier and joint-view simulation	Static corruption, one cohort and equal full sums
Combined observer	Joint business-view simulation	Same complete output and no authority signing key

4.1. Integrity and First Exposure

Table 5 separates the premises for history closure from those for observer privacy.

Definition 5 (Closure game)**.** The challenger initializes honest service keys, enrolled identities and genesis state. A bounded PPT adversary controls malicious clients and delivery. It submits packages, observes responses, replays, replaces, reorders or suppresses messages, and invokes admission and window closure. Honest signing keys and committed service state remain outside its control. The challenger records the locked P_w, R_w, invocation and response events, committed objects and receiver application counts. The adversary wins if the Decoder accepts a release for which no legal linearization exists, or any membership, application-count, common-object or reconstruction condition of Definition 3 fails, or if a native role exports a share that violates Eq. (6). Refusal without an unsafe export is not a win. The game fixes the strict profile. Write $\mathrm{Adv}_{\mathrm{closure}}(\mathcal{B}) = \Pr[\mathsf{Win}]$. Serializable durable state and honest service execution are premises of this game.

Lemma 1 (Native moment fidelity)**.** *Assume sound relation certification, commitment binding, authenticated common body identity, correct receivers and native validity. Let the certified canonical coordinate be* $\mathbf{x}_r$*, with* $p = \mathsf{Grid}(\mathbf{x}_r) < D < q$ *and* $b \in \{0,1\}$*. Let* $x = bp + (1-b)D$*, with canonical* $r_A, r_B, T \in [0, q-1]$*, and* $F(v) = \sum_{j=0}^{D} jv[j]$*. Suppose the retained outputs satisfy* $v_A + v_B = \mathbf{e}_y$ *and*

the receipts satisfy $m_S = F(v_S) + r_S$. *If certification fixes* $T = x + r_A + r_B \pmod q$ *and admission checks* $m_A + m_B = T$, *the reconstructed contribution after sentinel removal is* $b\mathbf{e}_p$.

Proof. Opening checks fix the certified masks. Linearity gives $y = F(v_A + v_B) = T - r_A - r_B = x \pmod q$. Since $x, y \in [0, D]$ and $D < q$, this is integer equality, and removing sentinel D yields $b\mathbf{e}_p$. □

Proposition 1 (First-exposure separation)**.** *Under the share-observation interface of Section 2.1, suppose two metadata-compatible admissible histories have the same aggregate on a full accepted set* $\mathcal{C}$, *but different aggregates on a permitted native batch* $R \subsetneq \mathcal{C}$. *Native validity, atomic admission and exact-set checking at final publication do not imply exposure-prefix safety or privacy relative to the full aggregate. This holds even for a wrapper that successfully publishes* $\mathcal{C}$ *on complete delivery.*

The separating execution delivers complementary shares of R to the corrupted role and Collector, then refuses publication because $R \neq \mathcal{C}$. Appendix A constructs the two legal histories. The exposure distinguishes the histories despite the later publication refusal.

Proof interface. The privacy simulation generates proofs for honest sources, while an adversary can reuse real or simulated proofs and combine material from related setup keys. A single-proof knowledge-soundness statement does not specify this source-reuse interface. Lemma 2 supplies the required alternative. An accepted proof yields a witness for a new statement or refers to a registered source with the same semantics. Lemma 1 then binds the contribution, and atomic history and durable export give the event induction in Appendix A.

Related-setup extraction. The analysis uses typed generic bilinear groups without a target-to-source map, correct setups with unknown trapdoors, independent latent variables and ceremonies with at least one honest contribution each, nonzero independent phase-two scalars, and finite source and oracle budgets. Correlated key points retain their setup relations. Only the current genesis and transition namespaces are accepted. Auxiliary keys remain visible to the adversary. The inventory and allowed source encodings are specified in Appendix B.

Statements are registered before fresh proof coins by key namespace, relation and ordered public signals. Honest-source reuse keeps its registered meaning. Known or extracted witnesses are cached, and malicious owners receive no witness-free proof service.

Lemma 2 (Related-setup extraction)**.** *Under these source rules, every accepted proof yields a witness or names a registered source with identical key, relation and ordered public signals, except with probability at most* g. *Appendix B gives its finite budget.*

Proof intuition. The joint source representation tracks real and simulated proofs under each key. Comparing the target verification equation separates registered target-source reuse from fresh statements. For fresh statements, generic bilinear group divisibility forces the proof elements to encode a valid assignment for the target relation, extracting the witness; unequal formal expressions are charged to the generic group collision bound. □

Atomic commits witness Definition 2. Durable receiver indexes enforce the application counts in Definition 3. Appendix A connects both invariants to export.

Theorem 1 (Accepted-history composition at first exposure)**.** *In Definition 5 under Section 2.1, let* B_C *bound the combined probability of whole-execution proof extraction, commitment binding, channel-package authentication, digest collision, and origin guessing failures, and let* v_N *bound native validity failure. Correct serialized durable execution is a premise. For the verification-key tag in* v_N, *the integrity instantiation adopts the SHA256 one-wayness assumption on uniform 128-bit keys with resource-bounded advantage* a_{OW128}. *Then*

$$\mathrm{Adv}_{\mathrm{closure}} \leq \min\{1, B_C + v_N\}. \tag{13}$$

Concrete parameterizations, query budgets, and algebraic expansions for B_C *and* v_N *appear in Appendix B. Outside those failure events, every first export and retry satisfies Eq.* (6) *with one immutable snapshot. The accepted log refines a legal ADWC history under Definitions 1 and 2. Every successful strict release satisfies Definition 3 and reconstructs*

$$\hat{\mathbf{h}}_w = \sum_{r \in I_w} b_r \mathbf{e}_{\mathsf{Grid}(\mathbf{x}_r)}. \tag{14}$$

Each record has one common admission object across its proof, consumed packages, committed transition, and release certificate. These invariants persist through recovery and collection failure.

Proof sketch. Under the source rules of Lemma 2, honest same-source reuse retains its registered semantics, malicious sources use known or extracted witnesses, and foreign namespaces or contexts are rejected. Commitment binding fixes the predecessor and anchor. Native validity and Lemma 1 bind its retained output to the proved cell. Induction over registration, lock, permit, commit and apply preserves the fixed candidate maps and a legal accepted history. The export transaction checks full completion and fixes the share before sending. Recovery preserves that transaction's snapshot. The roles apply the durable certificate idempotently without a cross-database atomic commit. After a correct freeze, complete membership is stable under monotone application, so an earlier incomplete snapshot can refuse but cannot authorize a smaller set. Consequently every exposure prefix satisfies Eq. (6), and reconstruction gives Eq. (14). Appendix A gives the event cases. □

Equation (14) also supplies the admitted-history numeric input to any downstream mechanism, without changing that mechanism's own privacy or output policy.

Bounding verification-key guessing from the public 96-bit tag in v_N requires SHA256 one-wayness rather than collision resistance alone (Appendix B).

4.2. Joint Privacy with Feedback

Definition 6 (Single-cohort privacy experiment)**.** Fix a finite public resource budget before setup. Sample a correct setup and fix the task, profile, keys and static corruption set before any honest measurement. The budget covers honest and adversarial work, including warmups, zero-bit reports, failed and unprepared reports, verification, oracle calls, bytes and retries. Execution stops at the budget. The environment supplies two complete legal histories H_0, H_1 for honest sources with the same public skeleton μ. It fixes each source's challenge inputs before its first report, while the adversary can adaptively request reports from those histories. Honest reports reach admission over an independent authenticated confidential channel according to the input-ready plan. Local witness and proof generation time is outside the online view.

For $X \in \{A, P, P{+}A\}$, the challenger selects a uniform hidden bit d, runs the protocol on H_d, and interacts with a bounded PPT adversary $\mathcal{B}$ with the following view. In A, the correct authority's view includes its business records, receipts, diagnostics, database and backups, and its ordinary authorization secrets. It has no role decryption keys or aggregate output. The observer may compute offline with its keys but cannot use them to issue extra authority signatures, change the configuration or bypass the state machine. In P, one actively corrupted Aggregator and the Collector share their state, decrypted inputs, peer messages and actual available Collector results. The corrupted role can fabricate messages and shares or abort. In $P{+}A$, that observer also sees the correct authority's business view, without its authorization secret or malicious issuance. The adversary can control corrupted owners and schedule, delay, replay or drop messages within the model. Each view includes its visible proofs, ciphertexts, commitments, responses, refusals, retries, aborts and modeled online times. These outcomes are not assumed to be part of μ.

For each record, let $y_i^{(d)} = b_i p_i^{(d)} + (1 - b_i)D$ denote its declared native index, where $p_i^{(d)}$ is its declared primary cell. For each allowed complete cohort R_w, the ideal interface returns the declared complete-cohort histogram $\mathcal{O}_d(R_w) = \sum_{i \in I_w} \mathbf{e}_{y_i^{(d)}}$, including sentinel D. Write $\mathcal{O}(R_w)$ when the world is clear; because a corrupted role or source can alter shares, $\mathcal{O}_d(R_w)$ sums declared indices rather than the Collector's reconstructed output. The A simulator receives no aggregate query and its two histories need not have equal histograms. The P and $P{+}A$ histories have equal declared complete-cohort histograms for every allowed complete cohort. Their simulators may query that full cohort once at the boundary where the first honest share becomes decryptable, even if collection later fails. This exposure-time interface returns no output before that boundary. Retries reuse the reserved plaintext without another query. No subset query is permitted. The game covers one visible selector for the fixed task and key.

The adversary outputs a bit in Exp_X^d. Its distinguishing advantage is $\mathrm{Adv}_X(\mathcal{B}) = |\Pr[\mathsf{Exp}_X^1(\mathcal{B}) = 1] - \Pr[\mathsf{Exp}_X^0(\mathcal{B}) = 1]|$.

Theorem 5 uses a private-prefetch variant of this interface. After durable reservation of a complete cohort, send preparation can query its output before the first honest share becomes decryptable. The answer remains private if preparation aborts without sending. The proof couples this early evaluation with exposure-time evaluation, preserving observable prefixes, including executions that never expose a share. Recovery retains the same cohort and cached answer, maintaining indistinguishability across aborted runs.

Theorem 2 (Joint privacy of admission and aggregation views)**.** *Fix the privacy experiment of Definition 6 with static corruption, the strict profile, and the source and programmable-oracle models in Appendix B. The authority view is simulatable without honest coordinates or an aggregate. The aggregation view, including any shared authority business view, is simulatable from the observer's inputs, μ and the permitted complete-cohort histogram. The distinguishing advantage of each observer X is at most $\mathrm{Adv}_X \le \min\{1, 2e_X\}$, where e_X composes the primitive bounds above, the hidden HPKE encapsulations and the native stream simulation. Theorem 4 gives the concrete form and a hybrid outline.*

The masked receipts provide the key link. Let c be the actively corrupted role and h the honest role. For an honest source report, the honest role computes $m_h = F(v_h^{\mathrm{orig}}) + r_h$ over its retained share, where $F(v) = \sum_j jv[j]$. Admission accepts paired receipts (m_h, z) if and only if $m_h + z = T$. Because $T = x + r_h + r_c \pmod q$ and native one-hot validity guarantees $F(v_h^{\mathrm{orig}}) + F(v_c^{\mathrm{orig}}) = x$, an arbitrary corrupt receipt z passes admission if and only if

$$\begin{aligned} z &= T - m_h = (x + r_h + r_c) - (F(v_h^{\mathrm{orig}}) + r_h) \\ &= F(v_c^{\mathrm{orig}}) + r_c \pmod q. \end{aligned} \tag{15}$$

Thus, whether z passes depends entirely on $F(v_c^{\mathrm{orig}}) + r_c$—quantities determined strictly by the corrupt role's own input share and mask.

Crucially, this refusal predicate establishes that malicious feedback cannot serve as a probing oracle under the joint distribution $(v_c^{\mathrm{orig}}, r_c, T, m_h, \mathsf{status})$. This invariance is not a heuristic appeal to the marginal uniformity of r_h. Rather, authenticated source registration binds the complete client package prior to role processing, locking v_c^{orig} and r_c to the immutable body. Idempotent retries and linearized commits evaluate against this same fixed predicate. Because x, v_h^{orig}, and r_h cancel out in Eq. (15), admission outcomes carry zero mutual information with honest trajectories.

For the simulator, uniform T determines the coupled honest receipt in Eq. (15) prior to learning the cell. For the authority alone, uniform T, U give $(m_A, m_B) = (U, T - U)$ without an aggregate query. These identities join the receipt simulation to the native preparation transcript and the same causal history state.

The simulator first classifies each request by its registered body and exact statement, including the key namespace and ordered public signals. Honest same-source reuse preserves registered business semantics and its native handle.

Malicious sources use known or extracted witnesses. Foreign namespaces or contexts are rejected. A changed honest body under the same statement requires a digest collision or an unauthorized signature. For a fresh statement, extraction supplies the committed openings, not the ciphertext plaintext. A ciphertext distinct from the protected challenge tuple is handled by legitimate decryption and native execution. This routing permits malicious sources without providing an oracle for an honest source's challenge.

For an honest source, the coupled receipt in Eq. (15) uses the same native share as the preparation transcript. For a malicious source, let y_j be its extracted or known claimed index and $w_{h,j}$ the actual successful output of the honest role. The simulator does not assume that these outputs satisfy the honest-source identity. When reservation permits first exposure, the single complete-cohort query $\mathcal{O}(R_w)$ gives

$$V_{h,R_w} = \mathcal{O}(R_w) - \sum_{j \in I_w^{\mathrm{adv}}} \mathbf{e}_{y_j} - \sum_{i \in I_w^{\mathrm{hon}}} v_{c,i}^{\mathrm{orig}} + \sum_{j \in I_w^{\mathrm{adv}}} w_{h,j}. \tag{16}$$

Here I_w^{hon} and I_w^{adv} partition the eligible report identifiers. Subtracting the known or extracted malicious-source declarations leaves the honest-source sum. For each honest source, $v_{h,i} = \mathbf{e}_{y_i} - v_{c,i}^{\mathrm{orig}}$. Subtracting the original corrupt-role shares therefore gives their honest-role aggregate over honest reports. Adding the actual honest-role outputs $w_{h,j}$ for malicious sources completes the joint view without subset queries or assuming that those sources follow their declarations. Retries reuse the reserved plaintext. The authority-only simulator uses the zero-query receipt law above.

Appendix B gives the Histogram kernel and reduction budgets for the joint simulation.

5. Evaluation

The evaluation examines trajectory impact, binding, first exposure, recovery and the cost of the enforced contract.

5.1. Setup and Comparison Contract

Datasets and workloads. Offline replay aligns 100 selected legal histories per dataset by relative step using three fixed actor permutations. Replay uses 300-second slots and a 100-m grid spanning 10 km. Appendix A specifies centers and trajectory boundaries.

1) *T-Drive [17].* GPS trajectories from Beijing taxis[1].
2) *Porto [18].* Trip trajectories from Porto taxis[2].

Implementation and setup.

1) *Protocol.* SHTPC uses Janus [19] with two-role Prio3Histogram under its DAP draft [5, 20]. The histogram has 10,000 spatial cells and a sentinel.
2) *Hardware.* Binding components and the Full/T pairs use Xeon Gold 6148 at 2.40 GHz. Components use Node 22.22.1 and snarkjs 0.7.4.
3) *Full/T workload.* Five paired runs use synthetic geometry, fresh identities and keys, 100 genesis proofs, six priming steps per owner and one eligible step per owner. Each measured release contains one nonzero eligible report per owner; the six preceding steps initialize history rather than contribute to that release. The synthetic batch has no assigned wall-clock sampling span. Client generation is configured for 16 workers, at most 64 queued bodies and at most four bodies per public drive call.

Baselines and controls.

1) *Binding alternatives.* Evaluation compares against direct in-circuit seed expansion and a Poseidon-based share-vector binding circuit [21].
2) *Trusted generation (T).* A plaintext-trusted baseline retains Prio3 report generation, preparation and aggregation, registration, freeze, durability and collection, but replaces application-layer state transition proofs and masked-moment checks with direct plaintext report synthesis from coordinates, state and randomness while retaining genesis proofs; signed configuration selects T receipts or Full proof and sidecar verification, and eight-client controls reject cross-mode or tampered inputs.
3) *SCIF and PEAR.* An acceptance bridge is exercised for SCIF [11] alongside an eight-report validity run for PEAR [12] under their respective aggregation interfaces.
4) *Offline oracle.* Replay compares independent composition with a strict-contract oracle that assumes detection.

Batch execution. The timed workload processes available inputs as a fixed-cohort batch. All candidate bodies are registered and locked before native preparation. The 300-second slots belong to the trajectory replay, not a processing deadline for the synthetic Full/T run.

Metrics.

1) *Replay.* Cohort-normalized L1 error and target-cell counts measure impact.
2) *Native controls.* Acceptance, release and recovery report observed outcomes.
3) *Component and system costs.* Binding uses three matched inputs, one logical CPU, fresh Node processes, one excluded warmup per relation and input, and five interleaved pairs per input. System endpoints are the client-generation-through-collection window and full session.

5.2. Real-Trajectory Impact and Native Controls

The evaluation uses three complementary workloads. First, $N = 100$ real-trajectory offline replay quantifies hotspot distortion under report substitution or omission. Second, $N = 8$ live Janus controls on empirical traces verify end-to-end service integration and attack refusal.

1. https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/
2. https://archive.ics.uci.edu/dataset/339

Third, $N = 100$ (700-transition) synthetic runs benchmark full-scale pipeline throughput and recovery (Sections 5.5–5.6). Because transition circuit constraints and Prio3 share processing are invariant to specific coordinate values within the legal grid (Table 6), synthetic legal trajectories exercise the exact same cryptographic and database execution paths as empirical traces.

Offline impact. Substituted reports can sharply amplify an apparent hotspot. In the offline replay, replacing 20 of 100 reports produces 0.38–0.40 cohort-normalized L1 error. Targeting the clean modal cell raises its count from 3 to 22–23 for T-Drive and from 2 to 21–22 for Porto. The strict-contract oracle withholds every tested window with nonzero substitution or omission, trading availability for integrity. An aborted window has no defined output error.

Native Janus controls. Janus runs on the first eight selected pressure histories from T-Drive and Porto release exact clean outputs and refuse both substituted reports in each 25% attack run before commit, with no incomplete-set export. There is one clean and one attack run per dataset.

Trajectory impact. Replacing 20 of 100 reports inflates modal hotspot counts from 2–3 to 21–23 (0.38–0.40 normalized L1 error) in offline replay, while eight-client Janus runs reject all substituted reports before commit.

5.3. Compact Semantic Binding

For a native-valid report for cell 42 whose proof authorizes cell 2312, the masked-moment check rejects without a commit, head advance or quota consumption, exercising Lemma 1.

The standalone masked-moment relation has 5,779 constraints at 10,001 bins, versus 79,002,390 for direct seed binding, a 13,670.60× ratio (Figure 4). It relies on native one-hot validity, and native moment evaluation still scans the vector in $O(D)$.

The constructed vector alternative uses blinded Poseidon folds over each role's share and context. Coordinate-wise reconstruction binds the shares to the authorized one-hot, including the sentinel. Hidden-state policy and ticket binding remain, while the receipt and public-signal interfaces change from 25 to 22 signals. Canonical decoding remains a receiver condition. The comparison evaluates transition circuits outside live Janus. Table 6 fixes the history, coordinates, contribution bit, context and both per-report native retained shares for each of three synthetic inputs. They cover nonzero contribution, zero contribution at the sentinel, and a legal move 3,299 m from its K-anchor under the 3,300 m cap. Setup verification precedes timing. fullProve includes witness generation. All 30 measured proofs and six excluded warmups passed independent verification. The transition components have 13,327 and 851,715 constraints, distinct from the standalone relations.

Binding cost. At 10,001 bins, masked-moment binding uses 5,779 standalone constraints and accelerates transition proving by 89.83–89.91× over Poseidon share-vector binding on matched inputs.

TABLE 6. SAME-INPUT TRANSITION COSTS AT 10,001 BINS, WITH FIVE PAIRS PER INPUT ON XEON GOLD 6148. M IS SHTPC MASKED-MOMENT BINDING. V IS CONSTRUCTED POSEIDON VECTOR BINDING [21]. RATIOS ARE PAIRED V/M MEDIANS.

Metric	Method	Nonzero	Zero	Near cap
fullProve median (s)	M	3.454	3.447	3.460
	V	310.415	310.741	310.925
Minimum (s)	M	3.444	3.429	3.450
	V	310.084	309.901	309.878
Maximum (s)	M	3.469	3.475	3.524
	V	311.296	311.963	312.015
Paired V/M		**89.91**	**89.90**	**89.83**
RSS median (GiB)	M	0.296	0.298	0.297
	V	13.916	13.870	14.142

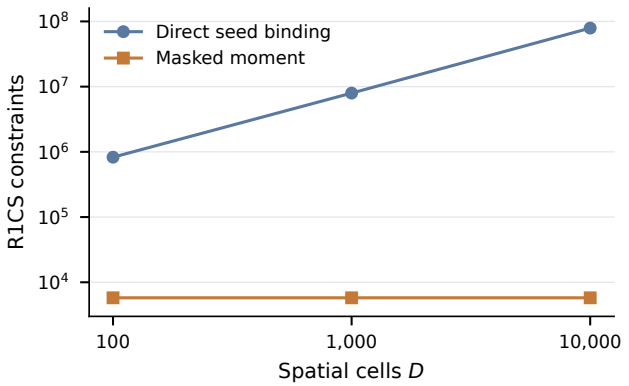


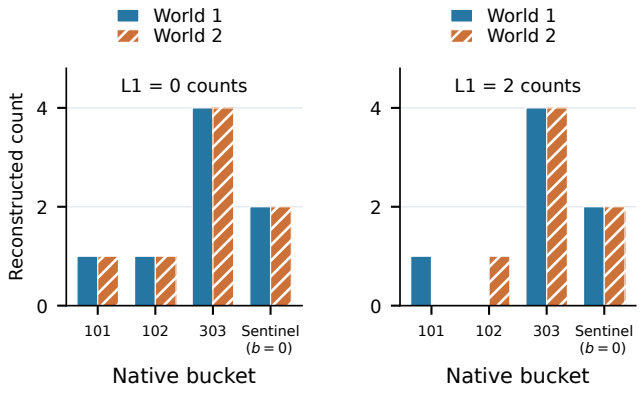
Figure 4. Standalone application binding R1CS sizes. Complete transitions and native $O(D)$ vector processing are outside this count.

5.4. Feedback and First Exposure

The feedback control shows why source authentication precedes receipt generation. With body-owner signature and authority-certificate checks disabled, copied honest reports with known masks produce receipts that reveal index 4237. A later admission refusal cannot erase that observation. The complete guard refuses the same input before decryption and feedback, while normal reports and exact retries continue through the protected path.

A complete aggregate can hide a difference that becomes visible in a subset. Two legal histories have the same eight-owner histogram but different seven-owner outputs (Figure 5). The native subset control exposes a two-count difference in unnormalized L1, the sum of absolute bucket-count differences. SHTPC refuses the incomplete set before an honest share is exposed, illustrating Proposition 1.

The external controls exercise different aggregation interfaces. The constructed SCIF acceptance bridge detects omitted positives, eligible zeros and equal-count substitutions after trusted reconstruction of an output without external member evidence [11]. A PEAR run reconstructs the exact eight-report, 10,001-bin output and rejects invalid inputs [12]. The control supplies neither an external accepted-



(a) (b)

Figure 5. Native control outputs for (a) the complete eight-member cohort and (b) the seven-member subset used for separation. The protected SHTPC path rejects the subset before exposure.

TABLE 7. NATIVE CONTROLS EXERCISING THE SECURITY OBLIGATIONS. ROWS DESCRIBE SEPARATE FIXTURES AND SCALES.

Boundary	Fixture / scale	Observed outcome
P1	Cell mismatch control	2312/42 mismatch rejected, no commit
P2	Source-feedback control	Guard-off revealed 4237, guard-on blocked
P3	Two-owner trace	Conflicting successor rejected, retry idempotent
P4	100 owners, 700 transitions	Missing 1 or 10 blocked, 100-member output restored
P4	Eight owners, seven exposed	Distinguishable seven-of-eight subset blocked

history head nor a locked cohort reference.

Feedback and first exposure. Source authentication blocks receipt-transplant leakage, and pre-export reservation prevents a distinguishing seven-of-eight subset exposure before any honest share becomes decryptable.

5.5. History Recovery

Durable redelivery restores the complete output after accepted decisions are withheld. In the 100-user campaign, withholding one or ten decisions delays release until redelivery restores all 100 eligible members and the exact 10,001-coordinate output. The campaign fixes 700 transitions. Affected services restart on retained databases while the key-holding controller remains live. Table 7 summarizes the observed obligations.

A two-owner native trace exercises whether retries and selective delivery preserve the committed history and frozen release set. The conflicting successor is refused, and redelivery restores the frozen set and exact output. The sequential trace complements recovery tests of concurrent preparation, changed peer bytes, arranged commit-first and freeze-first SQL orders, and failure paths that retain accepted state.

TABLE 8. SELECTED PHASE AND SESSION COSTS FOR FIVE FULL AND TRUSTED-GENERATION PAIRS.

Phase	Lifecycle time (s)	
	Full (SHTPC)	Trusted generation
History initialization median	3525.968	2504.718
Range	3498.93–3529.40	2492.19–2550.07
Of which setup and genesis	197.245	197.438
Eligible window median	548.266	381.117
Range	544.08–552.05	374.77–381.98
Total session median	**4351.280**	**3156.829**
Range	4323.88–4352.87	3149.91–3205.87

Setup and genesis are included in history initialization; window includes collection. Total session also includes harness initialization, resource snapshots, post-window observation and harness completion. Unlisted remainder medians are 277.245 s (Full) and 276.600 s (T). Phase medians are computed separately and need not sum to session median.

A finite TLA+/TLC 2.19 model complements these traces. Its 41 model configurations use one to three owners and reports, two roles, and one or two windows. Six safety and one fair-progress configurations pass, while guard removals expose invariant violations. Reachability and temporal controls exercise witnesses and fairness assumptions. The artifact specifies the case bounds and model.

History recovery. Durable redelivery restores exact 100-member outputs after 1 or 10 withheld decisions. Two-owner Janus traces and 41 bounded TLC configurations verify retry idempotence and conflict rejection.

5.6. Cost of the Enforced Contract

Table 8 reports five alternating Full/T pairs on the synthetic 100-client workload; all ten runs pass semantic, closure, metering and mode-control checks. Full completes the eligible batch in a median 548.266 s (versus 381.117 s for T) and the total fresh-history session in 4351.280 s (versus 3156.829 s for T), giving paired overheads of 1.44× on the eligible window and 1.38× on the session (paired T/Full medians 0.6958 and 0.7255), where T trusts plaintext report generation and omits transition proofs and moment checks. Within the eligible window for T, intermediate driver-phase observation medians are: report construction and registration 92.414 s, native drive 105.581 s, native freeze 41.403 s, and collection 61.555 s. Because these represent separately observed phase medians including driver synchronization, they cannot be directly summed as critical-path costs; the remaining unmeasured intervals and handoffs account for the rest of the window. Directly metered intervals are reported without conjecturing unmeasured disk sync or thread contention.

Comparing Full's 548.266 s window to T's 381.117 s gives a 167.149 s difference. Under ideal 16-way parallel client scaling, generating 100 transition proofs at 3.45 s each takes approximately 21.6 s ($100 \times 3.45/16 \approx 21.6$ s); thus, the difference also reflects server-side proof verification,

sidecar ingestion, statement registration, and receipt serialization. History initialization contains the per-run genesis and 600 priming transitions; these runs each start fresh on synthetic geometry to measure one eligible release without measuring multi-window recurring amortization.

Contract cost. Across five 100-client pairs, Full takes a median 548.3 s per eligible batch and 4351.3 s per session—1.44× and 1.38× plaintext trusted generation (T)—to verify hidden-state transitions and masked moments.

6. Related Work

To position SHTPC within the literature, related private aggregation systems classify along three core dimensions: (1) *Hidden-state history*: zero-knowledge verification of sequential per-identity state transitions (e.g., speed and anchor bounds); (2) $O(1)$ *native-output binding*: constant-size connection between client ZK claims and native secret-shared outputs (such as Prio3/VDAF) without in-circuit seed expansion; and (3) *First-exposure subset defense*: reconciling and freezing complete eligible cohorts before any honest share becomes decryptable. These dimensions delineate SHTPC's ecological niche: SCIF [11] enforces MPC input inclusion on memoryless inputs without client history policies or $O(1)$ native bridges; PEAR [12] evaluates static NP predicates without atomic history heads, constant-size native binding, or first-exposure locking; and Ball et al. [14] tracks global aggregate state rather than private per-client histories, leaving secret-share binding and subset exposure unaddressed. SHTPC uniquely addresses the intersection of all three dimensions.

Input validity, inclusion and verifiable aggregation. VDAFs and DAP [4, 5, 22, 23] decouple input validation from aggregation, extending to heavy hitters and streams [24–26]. ELSA, RoFL and fully linear PCPs enforce input validity on shared data [9, 10, 27], while VerSA and VeriFL verify aggregate correctness [13, 28–30]. Beyond input validity, SCIF [11] provides honest-input inclusion in MPC, and PEAR [12] evaluates zero-knowledge proofs for NP relations. SHTPC complements these systems by bridging zero-knowledge state proofs directly to native secret shares via masked one-hot index moments at $O(1)$ circuit cost, assuming correctly executing services for integrity.

Stateful workflows, credentials and membership. While stateful aggregation [14] updates global aggregate state, SHTPC verifies private sequential transitions for individual clients. Proof-carrying data [31, 32] propagates computation history, while Privacy Pass, DAP report authentication, zkcreds and ZEBRA authorize credentials [33–36]. In location services, VPriv, PrETP, PrivStats and ZKLP enforce charges, upload quotas or regional membership without revealing plaintext trajectories [1–3, 37]. However, static credentials do not prevent replay against stale state; SHTPC atomically authorizes successors against the actual per-identity history head. Furthermore, unlike federated aggregation schemes that tolerate dropouts via dynamic subsets [38–41], SHTPC enforces strict release: because partial subsets leak history privacy across distinguishing trajectories, it freezes complete eligible cohorts before first share exposure, using durable server redelivery for transport faults.

Output privacy. Differential privacy [42, 43] protects released heatmap statistics, complementing SHTPC's protocol-view protection.

7. Discussion and Conclusion

Hidden-state validation is incorporated into private aggregation. SHTPC binds each admitted state transition to the contribution retained by aggregation roles using native one-hot validity and a compact masked-moment relation, extended to complete-cohort first exposure via atomic admission and durable reservation. End-to-end Janus controls exercise these interfaces, while component benchmarks and lifecycle measurements quantify their costs.

The design targets coordinated sampling applications requiring complete cohort statistics with delayed batch release, preserving strict membership without silent dropout or intersection release, while durable server redelivery handles client disconnections. All guarantees apply to admitted claims under the stated dual trust model without establishing physical GPS truth or Sybil resistance. Differencing attacks on exact cohort totals require orthogonal output-level differential privacy. Future directions include streaming deployment, continuous windows, and stateful profiles beyond heatmaps.

8. Artifact Availability

The repository at https://anonymous.4open.science/r/semantic-artifact-4727/ contains code, proofs and data. Extended bounds, simulations, hybrids and multi-window proofs appear under artifact/proof/.

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Appendix A.
Interface, Policy and Composition Details

Interface and policy. An independent acceptance rectangle with valid pairs for two primaries also contains their mismatched cross-spliced pair. Thus every rectangle contained in the same-primary relation has one primary. For complete legal and illegal views X_0, X_1, let $a_i = \Pr[V(X_i) = 1]$. Completeness and soundness errors satisfy $e_c + e_s = 1 - a_0 + a_1 \geq 1 - |a_0 - a_1|$, which is one for identical views.
Trace preprocessing. For real-trace replay, local equirectangular projection uses radius 6,371,000 m and centers (116.4, 39.9) for T-Drive and (-8.61, 41.15) for Porto. Coordinates are shifted by 5,000 m and floored to integer meters. Points outside [0, 9999] in either coordinate are excluded from legal histories. The earliest observation per 300-second slot is used without gap filling. Pressure histories require eight consecutive occupied slots and, for Porto, one trip. The evaluated policy uses $K = 6$, a 40 m/s speed tier and a 3,300 m anchor cap.
Hidden-state policy. The hidden ADWC relation bounds squared displacement from the accepted predecessor by $\tau_u^2 \Delta t_t^2$, and from the anchor $\mathbf{a}_t = \mathbf{x}_{\max(0,t-K)}$ by $\min\{\tau_u^2 (\Delta T_t^{(K)})^2, c_{\text{policy}}^2\}$, with speed tier τ_u, anchor cap c_{policy} and $\Delta T_t^{(K)} = K \Delta t_{\text{slot}}$. Enrollment proves a hidden genesis. The first $\max(K, 2)$ accepted steps advance history without aggregation eligibility, and the authority checks the proof's commitment references against actual accepted history. Accepted steps, slots, timestamps and revisions are distinct.
First-exposure separation. Swap two owners' positive cells in metadata-compatible legal worlds. Complete aggregates coincide. A batch containing one swapped owner has different aggregates, so the corrupted role and Collector distinguish when the honest share arrives, proving Proposition 1.
Closure event induction. Registration stores full signed candidates without advancing history. Lock fixes P_w, R_w before private processing, and permits reference only actual committed predecessors and anchors. Commit atomically checks head and quota and records the successor, history and signed decision, preserving the legal transitions and real-time order of Definition 1. Apply changes $N_{S,w}(r)$ from zero to one only for eligible commits, with duplicate retries as no-ops. Reserve checks the registration projections in Eq. (12) and all record links, then stores an immutable selector and sum before send. Recovery preserves reservations. Pre-exposure abort retains partial commits and post-send failure retains the exposed snapshot. Induction gives Eq. (6) at every prefix, and native fidelity gives Eq. (14).
Export transaction refinement. The optimized export transaction refines Definition 3. By monotone role application, each eligible record transitions from pending to applied at most once ($N_{S,w}(r) : 0 \to 1$). Atomically validating the projections $\pi_{\text{reg}}(L_{S,w}) = P_w$ and $\pi_{\text{reg}}(M_{S,w}) = R_w$ against the locked registration graph ensures that the reserved share sum $\mathbf{a}_{S,w}$ incorporates all required members and no extraneous reports. Because the resulting snapshot is durably committed before network transmission, idempotent retries never modify the released cohort or authorize partial subsets.

Appendix B.
Joint Privacy and Concrete Bounds

Trust and lifecycle. Clients hold report-signing keys, admission enrollment and authorization keys, and roles input HPKE and receipt keys. The native verification key is uniform in $\{0,1\}^{128}$ and hidden from integrity-game clients. Only Collector holds the aggregate HPKE secret. Finalizer uses authenticated admission records, and Decoder the decrypted result and public verification keys.

Authenticated confidential registration binds bodies, permits, commits and receipts to owner, task, window and context. Labeled HPKE protects inputs, sidecars and Collector-bound shares. Decoder authenticates the finalized-set certificate. Durable journals and leases govern preparation and recovery.

Epochs bind signed context, while identity and quota survive rotation. Revocation, post-compromise recovery and forward secrecy are excluded. The view contains corrupted-role state, excluding honest-role secrets and backups. Durable data survive replay checks and recovery.

BabyJubJub Pedersen commitments [16] use canonical prime-order points, fixed public bases and uniform full-domain blinders. The hidden genesis and transition relations bind the true predecessor and K-anchor. With native field $q = 2^{128} - 7 \cdot 2^{66} + 1$, canonical $r_A, r_B, T \in [0, q-1]$ and $x \in [0, D]$, where $D < q$, give $0 \leq x + r_A + r_B < 3q$. The carry $\chi \in \{0,1,2\}$ therefore makes $x + r_A + r_B = T + \chi q$ an integer equality. Both sides are below $3q < 2^{130} < p_{\text{proof}}$, where the BN254 scalar modulus satisfies $2^{253} < p_{\text{proof}} < 2^{254}$. Thus equality in the proof field cannot wrap. Successful counts fit a signed 64-bit integer, so reconstructed histograms do not wrap modulo q.

Lemma 3 (Origin isolation)**.** *Outside authentication, digest and origin-allocation failures, a protected ciphertext enters private processing only in its registered source body.*

Proof. Allocation binds the identifier to its owner before publication. The complete body, encryption labels, installed lock and process permit fix the input before decryption. A transplant changes authenticated bytes or requires an already allocated identifier. Exact retries reuse the source and durable native state. □

Source family and extraction budget. Fix finite budgets before the challenge. All counts include honest construction, rejected inputs, offline queries, retries and rolled-back work. The inventory contains four keys over one phase-one setup (current genesis, current transition, historical genesis and N8) and two auxiliary $D = 10{,}001$ keys over another independent setup. All six phase-two scalars are nonzero and independent. No target public-wire point divided by its phase-two scalar is exposed; formal degree bounds across all setup keys ensure that polynomial identities hold over the field. Let $S = S_g + S_t + S_l + S_{\text{ext}}$ count fresh

TABLE 9. ADJACENT HYBRIDS AND THEIR LOSS. N COUNTS HONEST NATIVE SOURCES AND Q_i THE SEVEN STREAM BUDGETS.

Step	Change and simulator knowledge	Loss
1	Nonce deduplication over public unprepared sources	$N^2/2^{128}$
2	Initial usage, seed and binder programming	$N[Q_4 + 2(Q_3 + Q_6 + Q_7) + Q_5]/2^{128}$
3	Share and part programming with corrupt shares fixed	$N[Q_1 + Q_2 + 2Q_7]/2^{128}$
4	Prover-coin removal for unexposed seeds	$NQ_4/2^{128}$
5	Source simulation, full verifier, joint receipts, raw trace and first exposure	0
6	Strict-time truncation after $L + 128$ candidates	t_N

source proofs, and let $V = V_g + V_t + V_l$ count target-key verifications. The $Q_{G_1}, Q_{G_2}, Q_{G_T}, Q_{\mathrm{pair}}$ budgets count typed group-oracle outputs beyond sources, imports and verifications. The counts I_1, I_2 cover source-group imports. The proof-field ledger is

$$\begin{aligned} h_1 &= 65293168 + 2S + Q_{G_1} + I_1 \\ &\quad + 13V_g + 53V_t + 13V_l, \\ h_2 &= 16131831 + S + Q_{G_2} + I_2 + V, \\ h_T &= 8 + Q_{\mathrm{pair}} + Q_{G_T} + 7V, \\ d_G &= 8388612\tbinom{h_1}{2} + 4194303\tbinom{h_2}{2} + 12582915\tbinom{h_T}{2}, \\ g_{\mathrm{coll}} &= \min\{1, d_G/(p_{\mathrm{proof}} - 4194305)\}, \\ g_{\mathrm{aff}} &= \min\{1, S(3p_{\mathrm{proof}} - 2)/p_{\mathrm{proof}}^2\}, \\ g &= \min\{1, g_{\mathrm{setup}} + g_{\mathrm{coll}} + g_{\mathrm{aff}}\}. \end{aligned} \tag{17}$$

Here g_{setup} is the probability that setup sampling yields a zero phase-one evaluation point, α, β or phase-two delta, or an excluded evaluation root. The numerical example includes this term, $g_{\mathrm{setup}} \approx 1.9237 \cdot 10^{-70}$, under independent full-field sampling. For example, with $Q_{G_1} = Q_{G_2} = Q_{G_T} = Q_{\mathrm{pair}} = 2^{32}$, $S_g = S_t = S_{\mathrm{ext}} = V_g = V_t = 2^{19}$, $S_l = V_l = 0$, and $I_1 = I_2 = 2^{20}$, the stated setup and source assumptions give $g \le 2.6738 \cdot 10^{-50} < 2^{-150}$.

Native validity and key-tag assumption. Let Q_i count native stream queries by domain, in order measurement, proof share, joint randomness, prover randomness, query randomness, joint seed and joint part. Put $Q = \sum_{i=1}^{7} Q_i$, and let P_{att} count distinct attack preparation inputs. The actual public key tag is a 96-bit prefix of the verification-key hash. For G_5 external candidate-key queries, let K be the uniform 128-bit key and $Z = f(K)$ its tag. On the no-hit branch, each tag value determines at most G_5 candidate keys E_z. Averaging over tags gives

$$\Pr[K \in E_Z] = 2^{-128} \sum_z |f^{-1}(z) \cap E_z| \le 2^{96} G_5/2^{128}.$$

Thus $\kappa_{\mathrm{tag}} \le \min\{1, G_5/2^{32}\}$, without assuming uniform tag fibers. At $G_5 = 2^{31}$, the term $2\kappa_{\mathrm{tag}}$ already saturates the information-only bound. The factor two charges the two secret-key replacement hybrids. Primitive advantages remain symbolic. An additional SHA256 one-wayness assumption on uniform 128-bit inputs gives the alternative bound $\kappa_{\mathrm{tag}} \le a_{\mathrm{OW128}}$, obtained by scanning all queried keys with one additional hash per candidate. Collision resistance alone does not supply it. The native validity term is

$$\begin{aligned} v_N = \min\{&1, 10256(Q + P_{\mathrm{att}})/q + 2\kappa_{\mathrm{tag}} \\ &+ (2P_{\mathrm{att}} + Q)^2/2^{128}\}. \end{aligned} \tag{18}$$

Here $10256 = 10002 + 254$. The 10,001-bin invalid-input identity contributes at most $(10001+1)/q$ and a false gadget polynomial of degree at most 254 contributes $254/q$. The root-abort branch remains in the experiment. This invalid-input acceptance term is separate from the nonce-based privacy bound below, since a corrupted role already knows the verification key.

Exposure family.

Definition 7 (Exposure equivalence)**.** Using the declared indices of Definition 6, legal histories with common public structure μ are exposure-equivalent for R_w when

$$\sum_{i \in I_w} \mathbf{e}_{y_i^{(0)}} = \sum_{i \in I_w} \mathbf{e}_{y_i^{(1)}}. \tag{19}$$

Authority-only has no aggregate query and does not require this equality.

Proposition 2 (Effect of the exposure family)**.** *A fixed full cohort permits histogram-preserving reassignment. If every size-k subset can instead be exposed, $1 \le k < n$, equal sums on all such subsets force identical per-record native indices.*

Proof. Set $d_i = \mathbf{e}_{y_i^{(0)}} - \mathbf{e}_{y_i^{(1)}}$. For $k = 1$, each singleton immediately gives $d_i = 0$. For $2 \le k < n$, choose a common $k-1$ element set disjoint from i, j. Exchanging i, j in a size-k subset gives $d_i = d_j$. A size-k sum then gives $kd_i = 0$, hence $d_i = 0$ over integers. □

In the proof hybrid, full-domain blinding ensures language membership for any fixed legal reference history, with history-node aliases reusing the same point. Equation (15) gives the joint receipt law $(T, m_h) = (U, U - F(v_c^{\mathrm{orig}}) - r_c)$. For authority-only, uniform T, U give $(T, m_A, m_B) = (T, U, T - U)$.

The native model is a domain-separated programmable random oracle indexed by usage, seed, complete binders and raw offset. Repeated descriptors reuse rejected candidates and unconsumed prefetch too. Applying these bounds to the concrete TurboSHAKE128 XOF requires this idealization

beyond XOF pseudorandomness. The native kernel specializes the two-party, single-proof analysis in the 2024-09-25 revised full version of [23] to the pinned Histogram interface.

Lemma 4 (Concrete Histogram preparation view)**.** *The two-role, single-proof 10,001-bin Histogram has a 455-element proof, 202-element verifier and 128 interpolation points Ω. For valid inputs and query $t \notin \Omega$, its full verifier is $(0, W_0, \ldots, W_{199}, \sum_{j=0}^{99} W_{2j} W_{2j+1})$, with independent uniform W_i. Queries in Ω take the same root-abort branch in both worlds.*

Proof. Each valid-input wire has nonzero random-value coefficient $(t^{128} - 1)/(128(t-1))$. Its affine inverse makes evaluations independent uniform W_i, determining the gadget coordinate. For $t \in \Omega$ both worlds abort identically. □

Theorem 3 (Concrete two-party native privacy)**.** *In this programmable-oracle model, let N count honest native source reports, including unprepared reports. A single-world native stream simulation has loss at most*

$$b_N = \min\{1, (N^2 + NQ_*)/2^{128}\}, \tag{20}$$

where $Q_* = Q_1 + Q_2 + Q_5 + 2(Q_3 + Q_4 + Q_6) + 4Q_7$. Under Definition 6's exposure-time interface, the simulator queries one complete cohort when the first honest share becomes decryptable and none earlier, under Eq. (19)*.*

Proof intuition. Fix either corrupt role's original measurement/proof shares, key and nonce. Program fresh domain-separated streams, charging collisions and prior queries to Eq. (20). Sampling Σ from the verifier distribution and setting $z_h = \Sigma - z_c^{\text{orig}}$ makes the simulated honest share statistically identical to real execution; active responses z_c' are verified via $\mathsf{Decide}(z_h + z_c')$ and seed checks. Helper retries replay journaled bytes, while Leader recomputations preserve Σ, z_h and commit at most once, ensuring consistent simulation across retries without online verifier oracles. □

Theorem 4 (Joint privacy of admission and native release)**.** *Use Definition 6's privacy experiment, source and oracle models, and primitive bounds. Let X denote authority-only, active-role-and-Collector, or that aggregation view with authority business records. Let g be Eq.* (17)*'s bound, ϵ_H the full SHA256 collision advantage, a_X correct-party signature EUF-CMA advantages, and c_B Pedersen binding advantage. Let N_{id} count honest identifier draws and G_{id} adversarial guesses. With C_X hidden HPKE encapsulations and labeled IND-CCA advantages H_X, set*

$$B_X = \min\{1, g + \epsilon_H + a_X + c_B + N_{\text{id}} G_{\text{id}}/2^{128}\}, \tag{21}$$

$$e_X = \min\{1, 2(C_X + 10)B_X + H_X + b_N + t_N\}, \tag{22}$$

$$\text{Adv}_X \le \min\{1, 2e_X\}, \tag{23}$$

where $t_N \le \min\{1, 5N\binom{10129}{129}\rho^{129}\}$ and $\rho = (2^{128} - q)/2^{128}$. Authority-only needs no aggregate equality; the other two views require Eq. (19)*.*

Hybrid outline. Rows 1–4 of Table 9 sum to the bound in Eq. (20). The C_X labeled HPKE hops and ten macro hops each have two endpoints, each charged B_X. Their endpoint charges give $2(C_X + 10)B_X$, while H_X adds one IND-CCA advantage per hidden encapsulation, replacing honest ciphertexts and simulated receipts step by step. The five field streams cover measurement, proof share, joint, prover and query randomness; seed and part streams output bytes. Retries preserve the report, key, nonce, shares and randomness despite changing responses and verdicts. Repeated descriptors reuse their stream. Truncating after $L + 128$ candidates requires 129 rejections to fail; the union bound over $5N$ streams with $L \le 10001$ gives t_N. The two histories share one ideal view, so their distances add. □

Aggregation views have $C_X \le 2N$: one honest-role input and sidecar per honest source. Authority-only has $C_X \le 4N + E_{\text{enc}}$ for both roles and fresh visible aggregate encapsulations, including re-encrypted retries. Each H_X term uses one recipient-key-specific challenge with time, decryption-query and byte budgets; time includes reduction work. The integrity route uses B_C.

Theorem 5 (Predeclared multi-window privacy)**.** *Use Definition 6 with the predeclared k-window contract of Section 2.1 in one task, profile and key namespace. For P and $P{+}A$, require Eq.* (19) *for every declared complete cohort, including reserved unexposed cohorts. A joint simulator retains shared history, source, feedback and retry state. Persistent reservations fix each cohort and payload. Unlike Definition 6, aggregation-view simulators can privately prefetch after reservation, before the first honest share becomes decryptable, and cache through recovery. Reports remain single-window, while accepted-state references can cross windows.*

The authority-only analysis retains the same bound form, with resource budgets evaluated over the joint multi-window execution. For each observer,

$$e_X^{\text{MW}} = \min\{1, 2(C_X^{\text{MW}} + 10)B_X^{\text{MW}} + H_X^{\text{MW}} + b_N^{\text{MW}} + t_N^{\text{MW}}\},$$

and $\text{Adv}_X \le \min\{1, 2e_X^{\text{MW}}\}$. With joint native source count N_{MW}, $C_P^{\text{MW}} = C_{P+A}^{\text{MW}} = 2N_{\text{MW}}$ and $C_A^{\text{MW}} = 4N_{\text{MW}} + E_{\text{enc,A}}$, counting all fresh authority-visible aggregate encapsulations in $E_{\text{enc,A}}$. The deployed contract remains single-release per task and key.